What snapshot sampling does to order flow imbalance

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Abstract

We wanted to know what survives when order flow imbalance is calculated from increasingly sparse snapshots. On the same observed minute paths, summing transitions and applying the usual formula once to five-minute endpoints give opposite signs in 17.03% of material comparisons across 483 US stocks in 2019. Their top-20 absolute-value selections overlap by only 51.71%. High contemporaneous price fit does not resolve the difference: we show this analytically and with price-containing controls. We also examine two remedies. The tested endpoint regressions do not improve reconstruction of the minute sum. A price-aligned generalized-OFI construction reduces disagreement with its own sum from the standard measure's 21.71% to 7.02% on a matched four-date sample, but sufficient ten-level depth is available for only 57.60% of candidate windows. These results support a practical measurement audit and preserving contributions before coarsening. They do not establish event-level accuracy, a new trading signal, or a previously unknown general problem with snapshots.

Keywords: order flow imbalance; limit order book; temporal aggregation; measurement.
JEL classifications: C43, C58, G14.

1 The question we can answer with these data

This project began as an attempt to reproduce earlier order flow imbalance (OFI) results and look for useful predictive information. We had a large archive, but its finest observations were one-minute order-book snapshots. Most stocks had one book level; some had ten. More files could not resolve the basic limitation: the archive showed states of the book without the intervening sequence of orders, cancellations and executions.

That left a question we could answer directly. Given an observed minute path, how much changes when we retain only its five-minute endpoints? The OFI construction in Cont et al. (2014) sums signed book-event contributions. Summing contributions that have already been observed preserves that construction on the available grid. Applying the transition formula once to a longer interval's endpoints is a different operation. We compare the two on identical paths and with the same denominator.

The difference matters if it changes what a researcher selects or interprets. Our main finding is that endpoint substitution changes the sign in about one in six material five-minute comparisons. It also replaces roughly half the stocks selected by the largest absolute measurements. Ranking therefore does not make the choice of measurement harmless. This is a statement about feature selection, not about which portfolio would earn more.

The general limitation is already known. Su et al. (2021) explicitly discuss incomplete snapshots and use deeper queues to accommodate multi-tick price changes. Correct temporal aggregation is

also established practice: Ghysels and Nguyen (2019) sum tick-frequency snapshot contributions into minute OFI. Our current daily-OFI helper likewise sums weighted minute contributions. We have no basis for attributing our earlier strategy's weak results to endpoint substitution, or for claiming that prior OFI studies generally make this substitution incorrectly.

We contribute a paired audit of the measurement, its effect on rankings, and the limits of feasible remedies in this archive. We also show why current-return fit cannot certify measurement agreement. Multi-level OFI (Xu et al., 2019), cross-asset impact and lagged prediction (Cont et al., 2023), and static queue imbalance (Gould and Bonart, 2015) address related but different questions. Our results do not replicate or refute those designs. The proposed contribution is the combined empirical diagnostic, not a new definition of OFI or a new general theorem about information loss.

2 What the endpoints leave out

2.1 Two measurements on one path

Let $x_j = (b_j, q_j^b, a_j, q_j^a)$ denote best bid and ask prices and displayed sizes, with $0 < b_j < a_j$ and positive sizes. For one side, define

$$B(p_0, q_0; p_1, q_1) = q_1 \mathbf{1}\{p_1 \geq p_0\} - q_0 \mathbf{1}\{p_1 \leq p_0\}. \tag{1}$$

Write $B_j^b = B(b_{j-1}, q_{j-1}^b; b_j, q_j^b)$ and $B_j^a = B(-a_{j-1}, q_{j-1}^a; -a_j, q_j^a)$. Then $e_j = B_j^b - B_j^a$. Reversing ask prices gives the usual ask convention; a falling ask contributes negative pressure.

For n observed transitions, set $D_0 = (q_0^b + q_0^a)/2$ and compare

$$F_n = \frac{\sum_{j=1}^n e_j}{D_0}, \qquad C_n = \frac{B(b_0, q_0^b; b_n, q_n^b) - B(-a_0, q_0^a; -a_n, q_n^a)}{D_0}. \tag{2}$$

F_n is the minute sum and C_n the endpoint contribution. Neither is event-level OFI. The common positive denominator isolates the effect of dropping observations. Summing separately normalized minute contributions would instead introduce changing weights.

Proposition 1 (Aggregation error)**.** *For a one-sided observed path, the sum of transition contributions minus the endpoint contribution is*

$$\begin{aligned} E^B = \sum_{j=1}^{n} &\left(q_{j-1}\mathbf{1}\{p_j > p_{j-1}\} - q_j \mathbf{1}\{p_j < p_{j-1}\}\right) \\ &- q_0 \mathbf{1}\{p_n > p_0\} + q_n \mathbf{1}\{p_n < p_0\}. \end{aligned} \tag{3}$$

Consequently $F_n - C_n = (E^b - E^a)/D_0$, using negative prices on the ask side.

Proof. Checking the three price cases gives

$$B(p_{j-1}, q_{j-1}; p_j, q_j) = q_j - q_{j-1} + q_{j-1}\mathbf{1}\{p_j > p_{j-1}\} - q_j \mathbf{1}\{p_j < p_{j-1}\}.$$

The size differences telescope. Subtracting the same identity for the endpoints leaves (3). Subtract the reversed-ask identity and divide by D_0. □

Flat price paths have zero error even if sizes change. Monotone paths need not: repeated upward bid transitions count successive queues, while the endpoint calculation counts only the final queue. Price reversals are therefore unnecessary for a magnitude difference.

Proposition 2 (Endpoint ambiguity)**.** *There are valid paths with identical endpoint states and unnormalized net contributions $+q$ and $-q$, for any constant bid size $q > 0$. Every endpoint-only estimate has absolute error at least q on one of them.*

Proof. For $p > 2\delta > 0$, use bid prices $(p, p+\delta, p+2\delta, p)$ and $(p, p-\delta, p-2\delta, p)$. Keep bid size q, and hold a positive ask size and an ask price above $p + 2\delta$ fixed. Bid contributions are $(q, q, -q)$ and $(-q, -q, q)$; ask contributions vanish. Both endpoint contributions are zero. A common estimate u satisfies $2q \le |u - q| + |u + q|$. □

Repeating the closed path k times gives a kq lower bound when the number of observations is unrestricted. This is not an unbounded-error claim on a fixed five-minute grid. It also does not rule out useful conditional-mean estimates under additional assumptions. It rules out exact recovery from endpoints for all such paths.

2.2 Why a high price fit is not a check

Consider independent minute log returns $X_j \sim N(0, \sigma^2)$, $\sigma > 0$. Let the midpoint follow their geometric random walk, with bid and ask equal to $(1-s)$ and $(1+s)$ times the midpoint, $0 < s < 1$. Both sizes remain $q > 0$. Write $R_n = \sum_j X_j$.

Proposition 3 (Equal fit, different signs)**.** *Here $F_n = 2\sum_j \operatorname{sgn}(X_j)$ and $C_n = 2\operatorname{sgn}(R_n)$. Both have population contemporaneous OLS $R^2 = 2/\pi$ for every n, but*

$$\lim_{n\to\infty} \Pr(F_n C_n < 0) = \frac{\arccos\sqrt{2/\pi}}{\pi} \simeq 0.20595. \tag{4}$$

Both have zero covariance with an independent following-block return.

Proof. An upward move contributes $2q$ and a downward move $-2q$. With $S_n = \sum_j \operatorname{sgn}(X_j)$, symmetry and independence give

$$\operatorname{Cov}(R_n, S_n) = n\sigma\sqrt{2/\pi}, \qquad \operatorname{Var}(R_n) = n\sigma^2, \qquad \operatorname{Var}(S_n) = n.$$

Thus their squared correlation is $2/\pi$. Since R_n is Gaussian, $\operatorname{Corr}^2(R_n, \operatorname{sgn} R_n) = (\mathbb{E}|R_n|)^2/\mathbb{E}R_n^2 = 2/\pi$ as well. The joint central limit theorem for $(R_n/(\sigma\sqrt{n}), S_n/\sqrt{n})$ gives a bivariate standard normal limit with correlation $\rho = \sqrt{2/\pi}$. Its opposite-sign probability is $\arccos(\rho)/\pi$; the quadrant boundary has zero limiting probability. Independence gives the final covariance statement. □

Even finite n permits ties in F_n, whose probability vanishes in the limit. Simulations report ties separately: 100,000 paths with 1,001 increments give 20.52% disagreement and approximately 63.75% fit for each feature. This is an illustrative quote process, not a model of actual order arrival. Prices remain inside both features. The example shows that equally high explanatory fit cannot establish that they measure the same direction.

3 Data and design

The main sample contains 483 US stocks selected by 2017 coverage, observed through 2019: 378 have one-level source files and 105 have ten-level files. The main comparison uses level one for all stocks. There are 25,475,162 valid five-minute blocks across 358,807 stock-days. Blocks do

not overlap, start at 10:00 and end by 15:55. Shortened and exceptional sessions were excluded upstream. Quotes must be finite, positive and uncrossed throughout; absolute block log return must be below 0.2. These checks exclude 135 additional candidate blocks.

We examine one-, two-, five-, ten- and thirty-minute grids, plus a five-minute grid shifted by two minutes. The one-minute comparison is an identity check. A material comparison requires both $|F_n|$ and $|C_n|$ to exceed 0.1 starting-depth units. This cutoff followed a raw-file precision audit; smaller-cutoff results remain available. For each stock-day we divide opposite-sign material blocks by all material blocks, then average equally across stocks within dates and across dates, omitting empty stock-days. Pooled counts are reported separately.

Confidence intervals use 2,000 circular moving-block bootstrap draws with 20 trading dates per block and seed 279. Dates are resampled jointly across stocks. This preserves observed cross-stock dependence and within-block temporal dependence, but does not account for earlier research choices. Ten- and forty-date blocks provide sensitivity checks.

Each fitted model uses 2017 for fitting, 2018 to choose a ridge penalty from $\{0.01, 1\}$, and 2017–2018 for refitting before 2019 evaluation. Training data determine clipping at the 0.001 and 0.999 quantiles and standardization. The objective averages squared loss, penalizes slopes, and leaves the intercept unpenalized. Let L_m average squared errors within stock-days, across stocks within dates, then across dates. We report $R^2_m = 1 - L_m/L_0$ relative to the stock's training-mean forecast and gain $G_{m:r} = 1 - L_m/L_r$. Bootstrap draws recompute these loss ratios.

The same-block models explain current midpoint log returns. The forward baseline uses current and previous block returns, endpoint spread, log depth and queue imbalance, and trailing five- and thirty-minute returns. We add either F_n or C_n to predict the following block. Current and following absolute log returns must both be below 0.2; paired models use identical windows. Target-path validity is an eligibility filter, not a predictor. The chronological split does not make 2019 an untouched research holdout: these years had already been explored.

The extension protocol fixed three further comparisons before their new outcomes were inspected: stock-selection stability, endpoint reconstruction, and a bounded generalized-OFI benchmark. They remain exploratory. Fixed tanh and sign transforms were introduced after the initial fit results. Additional material-sample matching in the generalized comparison is also a post-result descriptive check. We retain these distinctions in the released protocols.

4 What changes in the observed sample

4.1 Signs and stock selection

At five minutes, balanced material sign disagreement is 17.03% (95% interval: 16.82–17.24%). Of 8,436,055 valid 2019 windows, 8,091,184 are material and 1,377,269 have opposite signs. The material filter retains 95.91%. The pooled ratio is not exactly the balanced estimate. Disagreement increases with the tested aggregation scales (Table 1).

The five-minute rates are 15.98% in 2017, 17.08% in 2018 and 17.27% on the shifted 2019 grid. All 483 stock-level 2019 rates are positive; the median is 17.64%, with tenth and ninetieth percentiles of 13.56% and 19.59%. One- and ten-level source groups give 16.77% and 17.97%, using level one in both. These are different stocks, so the contrast is not a depth effect. Cutoffs 0.001 and 0.01 give 17.91% and 17.80%; bootstrap lengths ten and forty give primary intervals of 16.87–17.22% and 16.79–17.26%.

Table 1: Material sign disagreement in 2019. Both absolute measures exceed 0.1; intervals use the date-block bootstrap.

Minutes	Opposite signs (%)	95% interval (%)
1	0.00	[0.00, 0.00]
2	11.79	[11.51, 12.09]
5	17.03	[16.82, 17.24]
10	19.49	[19.28, 19.68]
30	21.61	[21.36, 21.87]

We next rank stocks by $|F_n|$ and $|C_n|$ at each common five-minute timestamp in 2019. At least 100 valid stocks are required. Selection overlap is the intersection divided by 20, averaged over timestamps within dates and then dates. We repeat after dividing both measures by the same stock-specific scale: its 2017 ninetieth percentile of $|F_n|$. This reduces persistent cross-stock scale differences without using 2019 to set the scale. Ties follow a fixed universe order.

Table 2: Top-20 absolute-feature selection, 2019. Directional disagreement is measured among shared selections. These are feature ranks, not predicted returns or realized trades.

Scaling	Overlap (%; 95% interval)	Opposite signs (%)
Starting-depth units	51.71 [50.24, 53.20]	9.34
Common training scale	49.59 [48.09, 51.21]	8.76

The raw selections share about 10.34 of 20 stocks. Their absolute-rank correlation is 0.4215, so they remain related even though nearly half the selections change. Raw top-5 and top-50 overlaps are 54.66% and 47.42%; corresponding scaled overlaps are 54.28% and 45.92%. Ranking can remove a common magnitude rescaling. It cannot generally undo changes in ordering caused by discarded transitions. We do not infer a return loss from this turnover in selected names.

4.2 Explanation remains separate from forecasting

The frozen-size endpoint control

$$Z_n = \frac{q_0^b \operatorname{sgn}(b_n - b_0) + q_0^a \operatorname{sgn}(a_n - a_0)}{D_0} \tag{5}$$

retains price moves while removing within-block size changes. Its five-minute explanatory fit is 54.32%, compared with 24.25% for raw endpoint OFI. But bounding the latter with $\tanh(C_n)$ raises fit to 49.88% (Table 3). The scale of the transform is fixed at one. Much of the original gap therefore depends on functional form.

The frozen control still has 8.86% lower MSE than $\tanh(C_n)$, with descriptive interval 8.52–9.20%, on 8,436,055 matched windows. This compares nonlinear quote functions, not the causal contribution of queue updates. The return-sign row deliberately contains the outcome; it illustrates the limits of interpreting current-return fit as independent information.

The forward baseline has $R^2 = 0.0778\%$. Adding the minute sum gives 0.0732%, and adding the endpoint measure gives 0.0742%. MSE increases by 0.0046% and 0.0036%; relative-gain intervals are $[-0.0066, -0.0029]\%$ and $[-0.0053, -0.0021]\%$. These changes are precisely estimated but

Table 3: Same-block R^2 in percent, 2019. The lower panel consists of post-result descriptive checks. None is a forward forecast.

Feature	R^2	95% interval
Minute sum F_n	23.38	[21.23, 25.22]
Endpoint C_n	24.25	[22.19, 25.97]
Frozen-size endpoint Z_n	54.32	[53.28, 55.49]
$\tanh(F_n)$	37.45	[36.66, 38.38]
$\tanh(C_n)$	49.88	[48.84, 51.05]
$\mathrm{sgn}(F_n)$	32.31	[31.61, 33.12]
$\mathrm{sgn}(C_n)$	48.72	[47.69, 49.88]
$\mathrm{sgn}(R_n)$, outcome-based diagnostic	52.49	[51.46, 53.67]

economically tiny. Neither feature improves this baseline. The four primary loss contrasts retain significance under Holm correction, but that does not create a useful forward effect. We make no transaction-cost or execution claim.

5 What can be fixed, and at what cost?

5.1 Estimating the discarded sum from endpoints

An endpoint estimator might still approximate F_n statistically even though exact recovery is impossible over unrestricted paths. We test a ridge calibration of C_n and a nine-feature ridge model using C_n, Z_n, starting and ending queue imbalance, log-depth change, bid and ask changes divided by starting spread, $\tanh(C_n)$, and $|C_n|$. The target is the observed minute sum, not a return. Fitting follows the same chronological procedure.

On the same both-material sample as the headline, raw endpoint sign error is 17.03%, calibrated error is 17.04%, and the richer model's error is 18.93%. Across all eligible windows, their MSE changes relative to raw C_n are increases of 6.38% (95% interval: 2.66–11.41%) and 4.97% (1.37–10.08%). These particular regressions do not repair the measurement. Their failure is not a statistical lower bound for every possible decoder: distribution shift, clipping, regularization and model choice can all matter. We stop short of a broad model search because a better conditional forecast would still not identify the missing path.

5.2 Using deeper book information

The closest existing remedy is generalized OFI (Su et al., 2021). On an upward bid move, it counts new queues through the old best-price boundary and subtracts the old best size; a downward move reverses the construction. Unchanged best prices use the best-size difference. Ask prices are reflected as before. This uses more of the observed book and defines a different statistic.

We implement two conventions. The literal tick-grid version follows the one-cent contiguous-level interpretation of equations (9)–(10) in the 2021 preprint. The price-aligned adaptation sums occupied levels through the relevant price boundary; it agrees with the literal version on covered contiguous ladders and permits gaps between occupied levels. We return missing when the required

boundary lies outside the visible ten levels. We do not treat unavailable depth as zero. Neither implementation is log-GOFI.

The bounded sample uses all 105 ten-level stocks on the first regular available day of each 2019 quarter: January 2, April 1, July 1 and October 1. It contains 420 stock-days and 29,820 valid five-minute windows. For each convention, write G_n for summed minute generalized contributions and H_n for its endpoint contribution, both divided by D_0. We compare F_n with C_n and G_n with H_n on identical windows where all four absolute values exceed 0.1. This final matching rule is a post-result check. Rates average nonempty stock-days; four dates do not support a reliable temporal confidence interval.

Table 4: Generalized-OFI diagnostic on four dates. Coverage is the share of valid windows with both generalized constructions available, before the common material filter. Sign rates use that filter. Columns compare each operator with its own sum.

Convention	Coverage (%)	Material windows	Stock-days nonempty	Standard sign rate (%)	Generalized sign rate (%)
Price-aligned	57.60	16,264	414	21.71	7.02
Tick grid	32.03	9,222	234	21.22	3.47

On the price-aligned matched sample, disagreement falls from 21.71% to 7.02%. This is evidence of greater aggregation stability for a different operator on that sample. It is not evidence that H_n reconstructs standard F_n, or that either is closer to event OFI. The contiguous-grid result is more heavily selected: only 32.03% of valid windows have coverage and 234 stock-days have material comparisons. The two rows therefore cannot rank the conventions fairly against each other.

Selection also matters relative to the broad headline. Before the depth filter, standard disagreement on these four dates is 18.50%, not the full-year 17.03%. The result is a limited common-data comparison, not a replication of the Chinese-equity price-fit study. It suggests that deeper information can help, while making the coverage cost visible. The one-level majority cannot supply this remedy.

5.3 Preserving what has already been observed

When the fine observations still exist, the direct remedy is simpler. Calculate each transition first, then retain additive bid and reversed-ask sums before discarding the path. Store the observation spacing, boundary timestamps and states, and the quantities needed for the chosen denominator. Adjacent chunks merge by adding their summaries after checking the common boundary. This preserves observed OFI over unions of whole chunks, up to floating-point roundoff.

Our implementation rejects gaps, duplicate times, mixed spacings and inconsistent boundary states. It separately exposes starting-depth normalization and normalization by mean destination depth; the latter requires depth sum and transition count. Neither is the sum of individually normalized increments. A single chunk total also cannot answer arbitrary within-chunk queries or reconstruct a newly chosen time-weighting rule. Preserve finer increments or the specific weighted statistics when those uses matter. Asset identity and units must be consistent across each input stream.

This is ordinary additive aggregation, not a novel compression algorithm. If only coarse endpoints remain, exact repair is unavailable without extra assumptions. A data provider can prevent

this additional loss at collection time; a researcher can diagnose it before comparing studies or changing grids. Neither step recovers events missed by the original snapshot feed.

6 Checks, limits and interpretation

The original raw-file audit covered 18 stock-days across six stocks and all three years. Four exact zero flows became tiny nonzero cached values, with maximum sampled normalized error 0.000122. This motivated material thresholds; it is not a universal error bound. A separate all-stock replay reproduced every primary material count, and 210 saved models were checked against 7,560 stock-day losses. An omitted current-return cutoff in the forward mask affected six training windows for one stock across four grids. We corrected it and reran the affected models; the rounded five-minute conclusions did not change.

The extension adds formula and coverage tests, independent direct-sum and ranking replays on the four quarterly dates, and selected reconstruction refits. The released audit records distinguish full-sample checks from selected checks. These establish internal consistency, not event-level validity or novelty.

The archive may have survivorship and coverage selection. Three years of US equities need not represent other markets. The generalized comparison has only four dates and substantial censoring; the reconstruction search is deliberately limited. Research choices followed earlier work on the same archive. These limitations prevent a strong general claim about prediction or an optimal replacement for standard OFI.

We think the useful result is narrower. A change in the observation grid can change the measured direction and the stocks selected, while leaving a convincing current-price fit. Existing generalized constructions can reduce this instability, but their data requirements should be reported alongside the improvement. When possible, retaining contributions avoids the extra loss entirely. We would use this work to audit a data pipeline and qualify a measurement claim. We would not use it to redefine event OFI, dismiss earlier event-level findings, or claim that a profitable signal has been found. Whether this combined audit is sufficiently distinct for publication remains a question for expert review.

Reproducibility and data availability. The paper repository contains code, protocols, aggregate results, tests and audit records. It is private pending coauthor review; access is required. The shared research branch preserves earlier experiments. Raw minute quotes are not redistributed, so full empirical reproduction requires compatible source data. Synthetic checks and table verification need no raw data. This is a research draft pending coauthor review. AI assistance was used in analysis code, verification and drafting; listed authorship does not imply approval of this version.

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